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Multi-Currency Portfolio Optimization with General Risk Measures

Master's Thesis

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Chapter 5

Empirical Analysis

In this chapter, we analyze the out-of-sample performance of the joint and sequential overlay programs across the five risk measures and three universe sizes introduced in Chapters 3 and 4. In Section 5.1 we report the baseline backtests of the unregularized joint and overlay programs, together with the equally weighted benchmark. We show that the strategies trade excess return for lower volatility and drawdowns, that their risk-adjusted returns exceed those of the equally weighted portfolio in the medium and large universes but not consistently in the small universe, and that joint and sequential optimization produce similar realized performance across measures and universes. In Section 5.2 we repeat the analysis for the regularized joint and overlay programs and compare them against the regularized asset-only benchmarks, the fully hedged and unhedged portfolios. We show that regularization improves the Sharpe ratios, the most in the large universe, and that actively trading the currencies performs on par with full hedging. In Section 5.3 we consider the mean-risk formulation of the optimizers and show that adding a mean term substantially improves the risk-adjusted returns. We further show that the deviation measures benefit from the addition of a mean term more than the tail measures. Across the three considered backtests, the only metric where joint optimization consistently outperforms sequential optimization is volatility, i.e. risk reduction, where the joint program achieves lower realized volatility across all risk measures and universe sizes. This motivates the analysis of Chapter 6, where we focus on comparing the optimization approaches in terms of their risk reduction.

5.1 Baseline Backtest Results

We first report the baseline backtests of the unregularized Joint and Overlay optimizers across the five risk measures and three universe sizes considered in the thesis. We compare them against the equally weighted portfolio which serves as the no-optimization benchmark.

For the small, medium and large universes, we report in Table 5.1, the out-of-sample annualized excess return, annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, average asset turnover and average currency turnover of the Overlay and Joint optimizers and of the equally weighted portfolio. All quantities are computed on the out-of-sample excess return series produced by the backtest configuration presented in Section 4.3. We also report in Figures 5.1 to 5.3 the cumulative out-of-sample returns of the Joint optimizers together with the equally weighted benchmark, and in the lower panel of each figure the cumulative difference between the Joint and Overlay optimizers for each risk measure. The latter helps us visualize the realized return difference between the two optimization approaches over time.

Baseline Backtest Performance across Universe Sizes

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	1.77	1.12	1.25	0.67	0.61	1.26	1.39	0.61	0.62	0.69	0.19
Vol (%)	13.96	7.97	7.73	8.15	7.84	7.87	7.71	8.18	7.97	9.17	8.72
MD (%)	54.77	28.23	27.92	32.17	32.63	27.67	27.31	29.31	28.31	34.43	35.38
SR	0.13	0.14	0.16	0.08	0.08	0.16	0.18	0.08	0.08	0.07	0.02
SOR	0.17	0.19	0.22	0.11	0.11	0.22	0.25	0.10	0.11	0.10	0.03
Turnover^a	0.30	0.87	0.82	0.99	0.97	1.10	1.06	1.30	1.27	1.38	1.47
Turnover^c	0.92	1.70	1.88	2.13	2.38	1.51	1.84	2.39	2.98	2.66	3.46
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.09	2.41	2.53	2.37	2.54	2.75	2.65	3.16	2.95	2.39	2.70
Vol (%)	12.50	5.17	5.09	5.27	5.16	5.73	5.56	5.68	5.53	5.86	5.64
MD (%)	55.75	35.85	34.90	37.88	36.71	35.93	36.94	38.59	38.84	35.05	33.09
SR	0.33	0.47	0.50	0.45	0.49	0.48	0.48	0.56	0.53	0.41	0.48
SOR	0.44	0.62	0.67	0.60	0.66	0.69	0.68	0.75	0.73	0.55	0.66
Turnover^a	0.27	1.05	1.05	1.25	1.26	1.66	1.68	1.78	1.79	2.13	2.23
Turnover^c	0.92	1.16	1.37	1.20	1.36	0.98	1.25	0.97	1.67	1.02	1.85
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.26	2.74	2.79	2.73	2.76	2.44	2.40	3.28	3.44	2.39	2.46
Vol (%)	11.66	4.24	4.19	4.26	4.20	4.63	4.57	4.57	4.55	4.67	4.66
MD (%)	56.26	26.72	26.37	28.74	28.29	31.25	31.42	31.32	30.46	31.29	30.31
SR	0.37	0.65	0.66	0.64	0.66	0.53	0.52	0.72	0.76	0.51	0.53
SOR	0.49	0.85	0.88	0.84	0.87	0.70	0.70	0.95	1.01	0.68	0.70
Turnover^a	0.28	1.09	1.09	1.34	1.36	1.76	1.77	1.89	1.91	2.15	2.18
Turnover^c	0.91	1.06	1.24	1.07	1.21	0.97	1.16	0.97	1.43	0.99	1.62

Table 5.1: Performance metrics of the baseline backtests in the small, medium and large investable universes

Table 5.1 reports out-of-sample performance metrics for the unregularized optimizers in the three investable universes, with portfolios rebalanced quarterly. Panel A covers the small, Panel B the medium and Panel C the large investable universe. Each panel reports the unregularized sequential currency overlay (Overlay) and joint (Joint) optimizers, together with the equally weighted portfolio (EW). Columns are grouped by the risk measure that is minimized, that is, variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES) and second-order Expected Shortfall (SOES). The realized out-of-sample performance metrics shown are the annualized excess return (ER), volatility (Vol), maximum drawdown (MD), the Sharpe ratio (SR), the Sortino ratio (SOR), and average asset and currency turnover, respectively. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. More details on the optimization approaches, backtest configuration, and performance metrics are provided in Chapters 3 and 4.

Across all five risk measures and universe sizes, the Overlay and Joint optimizers achieve substantially lower volatility than the equally weighted portfolio. In the small universe, their volatility ranges from 7.71% to 9.17%, against 13.96% for the equally weighted portfolio, a reduction of 34% to 45%. In the medium and large universes, the reduction is even larger, by up to 59% and 64%, respectively. The equally weighted portfolio ignores the dependence between asset returns. Since it already holds a broad cross-section of stocks, most idiosyncratic risk is diversified away, and further adding stocks does little to reduce the remaining market risk,

which can only be addressed by taking the correlation structure of the assets and currencies into account. The optimizers instead tilt the portfolio toward assets with low risk, and a larger universe offers more such assets to choose from. As a result, maximum drawdowns are also substantially reduced, by up to 27 percentage points in the small, 23 in the medium and 30 in the large universe, relative to drawdowns of 54.77% to 56.26% for the equally weighted portfolio. Large drawdowns occur in market crashes, when the correlations between assets rise and diversification offers less protection. The optimizers reduce the portfolio's exposure to these crashes, but cannot completely mitigate them, since in a crash even low-risk equities tend to fall together, which is why the drawdowns remain sizeable for all strategies.

For excess returns, we observe the opposite effect. The Overlay and Joint optimizers deliver lower excess returns than the equally weighted portfolio for every risk measure and universe size. This is partially due to the fact that the optimizers are minimizing risk without any explicit return term in the objective function. We address this in more detail in Section 5.3. Secondly, since much of the volatility reduction achieved by the risk minimizing portfolios comes from a lower exposure to the common movement of the equity markets (i.e. lower market beta), the optimizers also earn a smaller share of the equity risk premium. But excess returns alone say little about performance when the portfolios carry different levels of risk, so we compare them on a risk-adjusted basis.

In the small universe, the Sharpe ratios of the optimizers range from 0.02 to 0.18, against 0.13 for the equally weighted portfolio, and only the variance and mean absolute deviation optimizers improve on the benchmark. In the medium and large universes, every optimizer outperforms the equally weighted portfolio, with Sharpe ratios between 0.41 and 0.56 against 0.33, and between 0.51 and 0.76 against 0.37, respectively. In the small universe, the optimizers lose more excess return than they reduce risk, while in larger universes they can diversify across more low-risk stocks, and the reduction in volatility outweighs the lost return. Sortino ratios follow the same pattern, so accounting only for downside risk does not change the comparison.

Both the Overlay and Joint optimizers have much higher asset and currency turnover than the equally weighted portfolio. This is expected since the equally weighted portfolio has fixed target weights and only trades to undo the price drift. Comparing the asset turnover of the Joint and Overlay optimizers, we see that it is very similar, differing at most by 6%. This tells us that the asset weights are determined primarily by the risk measure being optimized and not by the optimization approach itself. The difference between the two optimizers is in currency turnover, where Joint is higher than Overlay in every measure-universe combination, by between 11% and 82%. Since returns are reported net of transaction costs, the comparisons above already incorporate these differences. The extra currency turnover does not reduce the risk-adjusted returns of Joint relative to Overlay, and the higher turnover of both optimizers does not change their advantage over the equally weighted portfolio.

Finally, comparing the performance of the unregularized Joint and Overlay optimizers under the same risk measures, across all reported metrics, we see that they perform very similarly. The Joint optimizer does consistently achieve a lower volatility, but the differences shrink as we increase the universe size, from up to a 7% decrease in the small universe, to essentially no difference in the large universe. This is in line with the joint program being able to exploit the asset and currency dependence structure when the number of assets compared to the number of currencies is smaller. We study the risk reduction performance and mechanism in more detail in Chapter 6. The other metrics differ by small amounts, with excess returns differing by at most 0.5 percentage points, maximum drawdowns by at most 1.96 percentage points and Sharpe ratios by at most 0.07. However, the sign of these differences changes across risk measures within every universe, so one optimization approach does not consistently

outperform the other.

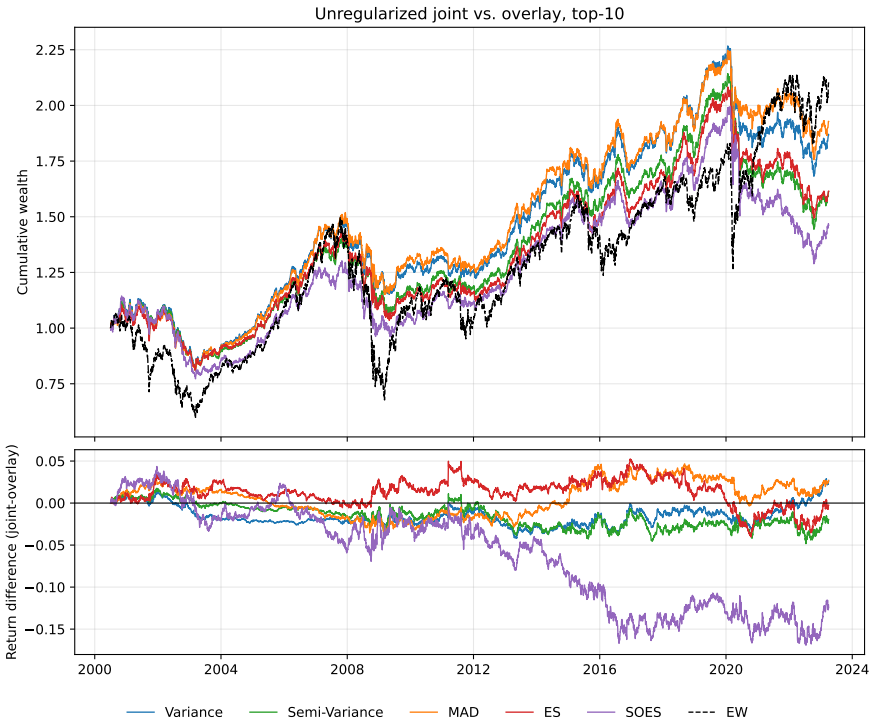


Figure 5.1: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy).

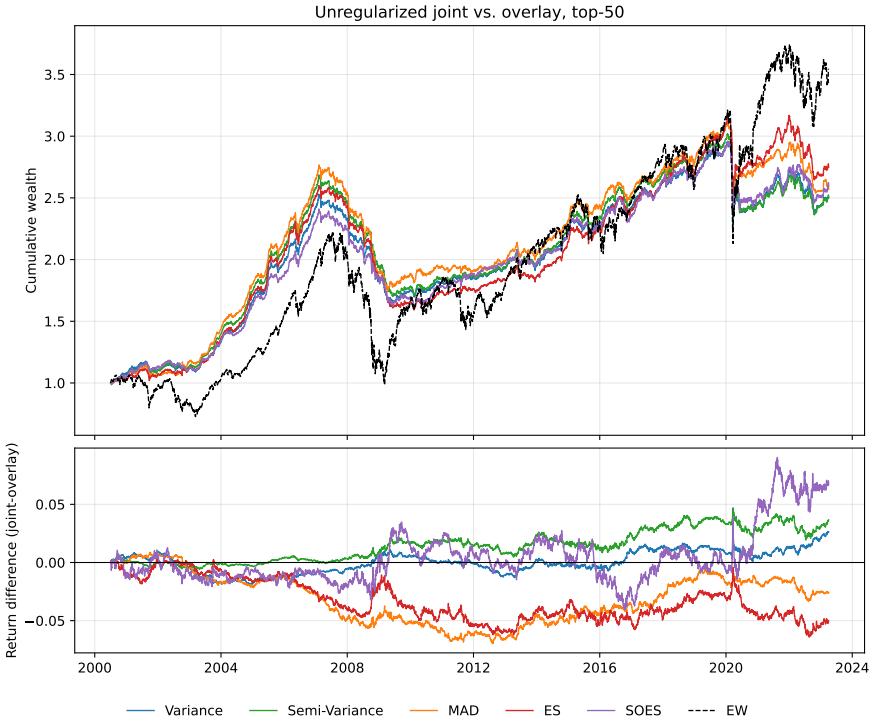


Figure 5.2: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy).

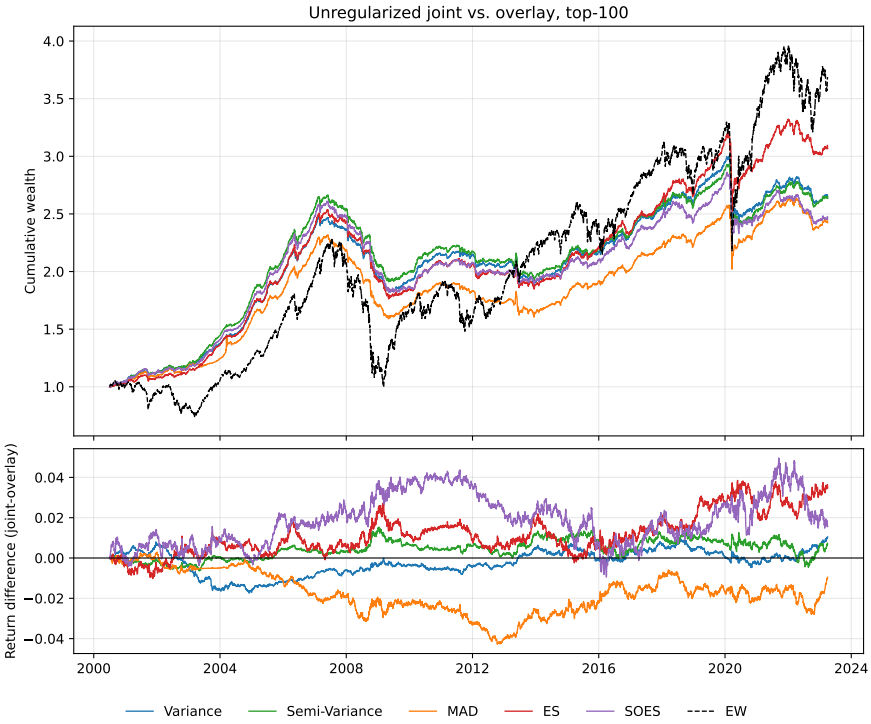


Figure 5.3: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy).

Figures 5.1 to 5.3 show the out-of-sample performance of the unregularized optimizers in the small, medium and large investable universes, respectively. In each figure, the top panel shows the cumulative wealth of the joint optimizers and the equally weighted portfolio, and the bottom panel shows the cumulative sum of the daily return differences between the joint and the overlay optimizers. The portfolios are rebalanced quarterly and reported net of transaction costs. The corresponding performance metrics are reported in Panels A, B and C of Table 5.1, respectively.

Figures 5.1 to 5.3 present the cumulative performance of the Overlay and Joint optimizers alongside the equally weighted benchmark. In each universe, the top panel shows that the Joint optimizers perform better than the equally weighted portfolio during the first major market crashes in the sample, the dot-com crash and the global financial crisis. This advantage is particularly evident in the medium and large universes during the global financial crisis, where the optimizer strategies exhibit a much smaller drawdown compared to the equally weighted portfolio. The equally weighted portfolio recovers quickly after the COVID-19 market shock and ultimately attains the highest terminal wealth across all three universes, albeit at considerably higher volatility and with deeper drawdowns throughout the sample. In contrast, the Overlay and Joint optimizers follow more stable growth paths, consistent with their lower realized volatility reported in Table 5.1.

The bottom panels show the cumulative daily return difference between the Joint and Overlay optimizers for each risk measure, and confirm that the two optimization approaches track each other closely throughout the sample in terms of returns. The cumulative differences oscilate around zero in every universe size and for every measure, with the exception of second-order Expected Shortfall in the small universe, where the Joint optimizer drifts persistently below the Overlay optimizer from the early 2010s onward.

Overall, we see a consistent pattern across all the considered risk measures. The Overlay and Joint optimizers reduce volatility and drawdowns substantially relative to the equally weighted portfolio. The Sharpe comparison depends on the universe size. In the small

universe, the Joint and Overlay optimizers do not consistently outperform the equally weighted portfolio, while in the medium and large universes they do. The Joint and Overlay optimizers produce very similar out-of-sample metrics across all risk measures and universe sizes, with Joint consistently delivering lower volatility and higher currency turnover but no consistent advantage on risk-adjusted returns.

5.2 Regularization

We now repeat the analysis of Section 5.1 for the regularized versions of the Joint and Overlay optimizers. We add L_1 and L_2 penalties on the asset weights and, separately, on the currency weights to the objective, following the regularization scheme of Chapter 3. The penalty weights are selected at every rebalancing date by a grid search over the validation window described in Section 4.3. All other aspects of the backtests remain unchanged. We also compare the Joint and Overlay optimizers against the regularized Assets-Full (Full) and Assets-Zero (Zero) optimizers which serve as benchmarks for optimizing only the asset weights under the same risk measures, with the currency weights either fully hedged or set to zero, respectively. Table 5.2 reports the change in the Sharpe ratio between the unregularized and regularized optimizers, and Tables 5.3 to 5.5 report performance metrics of the regularized optimizers and the three benchmarks.

Change in Sharpe Ratio from Regularization

	Overlay			**Joint**		
Risk Measure	**Small**	**Medium**	**Large**	**Small**	**Medium**	**Large**
Variance	+0.02	−0.02	+0.12	0.00	−0.05	+0.10
Semi-Variance	+0.05	+0.02	+0.19	+0.03	−0.03	+0.18
MAD	+0.04	−0.02	+0.02	+0.03	+0.01	+0.04
ES	+0.05	−0.01	+0.15	+0.09	+0.02	+0.13
SOES	+0.04	+0.12	+0.38	−0.02	+0.08	+0.38
Average	+0.04	+0.02	+0.17	+0.03	+0.01	+0.17

Table 5.2: Change in the out-of-sample Sharpe ratio of the Overlay and Joint optimizers from regularization

Table 5.2 reports the change in the out-of-sample Sharpe ratio from regularization, computed as the Sharpe ratio of each regularized optimizer in Tables 5.3 to 5.5 minus that of its unregularized version in Table 5.1, for the Overlay and Joint optimizers in the small, medium and large investable universes. Rows correspond to the risk measure that is minimized, and the last row reports the average change across the five risk measures.

We see in Table 5.2 that regularization improves the Sharpe ratios of the Overlay and Joint optimizers for most risk measures and universe sizes. The improvement is the largest in the large universe, where the Sharpe ratio rises by 0.17 on average, while the effect is modest in the small universe and mixed in the medium universe. The penalties stabilize the weights by limiting the extreme positions that the unregularized programs fit to noise in the lookback window, which becomes more important when the number of assets increases relative to the length of the lookback window. The unregularized second-order Expected Shortfall optimizers are the least stable, since the measure is driven by the squared losses of the few most extreme observations in the lookback window. Accordingly, regularization benefits them the most, raising their Sharpe ratios by 0.38 in the large universe.

Regularized Backtest Performance: Small Universe (top 10 stocks per economy)

Panel A1: Regularized Overlay and Joint Optimizers

	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	1.29	1.24	1.05	0.84	1.60	1.64	1.07	1.31	0.98	0.03
Vol (%)	7.82	7.76	7.91	7.72	7.81	7.75	7.99	7.85	8.52	8.64
MD (%)	27.81	27.80	30.63	30.43	27.83	26.01	29.96	28.89	36.02	36.22
SR	0.16	0.16	0.13	0.11	0.20	0.21	0.13	0.17	0.11	0.00
SOR	0.22	0.22	0.18	0.15	0.28	0.29	0.18	0.23	0.16	0.00
Turnover^a	0.93	0.96	1.09	1.01	1.04	1.06	1.07	1.17	1.40	1.59
Turnover^c	1.43	1.47	1.72	1.77	1.36	1.44	1.70	1.77	2.04	2.12

Panel B1: Benchmarks

		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	1.77	1.05	2.64	1.08	2.65	1.66	3.13	0.85	2.44	0.55	1.93
Vol (%)	13.96	8.39	11.80	8.41	11.72	8.36	11.70	8.51	11.84	9.12	12.35
MD (%)	54.77	28.00	42.18	29.21	41.12	28.59	40.88	30.81	44.28	33.06	44.84
SR	0.13	0.12	0.22	0.13	0.23	0.20	0.27	0.10	0.21	0.06	0.16
SOR	0.17	0.17	0.30	0.17	0.31	0.27	0.36	0.13	0.28	0.08	0.21
Turnover^a	0.30	1.02	1.02	1.16	1.17	1.08	1.08	1.27	1.23	1.66	1.67
Turnover^c	0.92	0.93	0	0.93	0	0.92	0	0.93	0	0.94	0

Table 5.3: Performance metrics of the regularized optimizers in the small investable universe (top 10 stocks per economy)

Regularized Backtest Performance: Medium Universe (top 50 stocks per economy)

Panel A2: Regularized Overlay and Joint Optimizers

	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	2.38	2.35	2.41	2.33	2.83	2.95	3.16	3.08	2.91	3.10
Vol (%)	5.26	5.19	5.16	5.08	6.16	6.08	5.74	5.58	5.52	5.52
MD (%)	35.03	34.25	36.39	34.57	35.93	35.52	38.57	36.38	30.41	30.20
SR	0.45	0.45	0.47	0.46	0.46	0.49	0.55	0.55	0.53	0.56
SOR	0.59	0.59	0.62	0.61	0.75	0.79	0.77	0.76	0.71	0.76
Turnover^a	1.14	1.13	1.24	1.27	1.31	1.32	1.53	1.57	1.56	1.97
Turnover^c	1.11	1.11	1.18	1.17	1.09	1.06	1.26	1.30	1.39	1.32

Panel B2: Benchmarks

		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	4.09	2.52	3.86	2.55	3.92	2.94	3.84	3.26	4.54	3.02	4.48
Vol (%)	12.50	5.31	9.58	5.22	9.43	6.19	9.97	5.84	9.87	5.60	9.78
MD (%)	55.75	35.05	42.05	36.53	41.31	35.90	43.81	38.54	42.37	29.76	36.74
SR	0.33	0.48	0.40	0.49	0.42	0.48	0.38	0.56	0.46	0.54	0.46
SOR	0.44	0.62	0.55	0.65	0.57	0.77	0.57	0.78	0.64	0.72	0.63
Turnover^a	0.27	1.07	1.07	1.34	1.35	1.30	1.30	1.60	1.60	1.81	1.81
Turnover^c	0.92	0.95	0	0.95	0	0.96	0	0.95	0	0.95	0

Table 5.4: Performance metrics of the regularized optimizers in the medium investable universe (top 50 stocks per economy)

Regularized Backtest Performance: Large Universe (top 100 stocks per economy)

Panel A3: Regularized Overlay and Joint Optimizers										
	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	3.33	3.25	3.56	3.52	2.72	2.73	3.88	3.91	4.05	4.09
Vol (%)	4.35	4.29	4.27	4.19	4.97	4.85	4.45	4.40	4.57	4.52
MD (%)	26.27	26.40	27.33	27.29	31.28	31.63	30.61	30.11	25.84	24.70
SR	0.77	0.76	0.83	0.84	0.55	0.56	0.87	0.89	0.89	0.91
SOR	0.99	0.98	1.09	1.10	0.73	0.75	1.14	1.17	1.17	1.20
Turnover[a]	1.14	1.10	1.40	1.33	1.60	1.56	1.42	1.64	1.65	1.87
Turnover[c]	1.06	1.03	1.06	1.06	1.01	1.03	1.15	1.23	1.23	1.26

Panel B3: Benchmarks											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	4.26	3.16	4.40	3.56	4.94	2.83	4.13	4.00	5.28	3.75	4.92
Vol (%)	11.66	4.37	8.99	4.28	8.92	4.94	9.36	4.46	9.21	4.57	9.28
MD (%)	56.26	26.60	33.06	27.84	34.27	31.28	36.99	31.60	35.12	30.41	35.50
SR	0.37	0.72	0.49	0.83	0.55	0.57	0.44	0.90	0.57	0.82	0.53
SOR	0.49	0.93	0.68	1.09	0.77	0.77	0.62	1.18	0.80	1.09	0.74
Turnover[a]	0.28	1.10	1.10	1.35	1.35	1.59	1.59	1.61	1.62	1.84	1.84
Turnover[c]	0.91	0.96	0	0.97	0	0.97	0	0.96	0	0.97	0

Table 5.5: Performance metrics of the regularized optimizers in the large investable universe (top 100 stocks per economy)

Tables 5.3 to 5.5 report out-of-sample performance metrics for the regularized optimizers in the three investable universes, with portfolios rebalanced quarterly. Panels A1, A2 and A3 report the regularized sequential currency overlay (Overlay) and joint (Joint) optimizers, and Panels B1, B2 and B3 the benchmarks, the equally weighted portfolio (EW) and the asset-only optimizers Assets-Full (Full) and Assets-Zero (Zero). Columns are grouped by the risk measure that is minimized, that is, variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES) and second-order Expected Shortfall (SOES). The realized out-of-sample performance metrics shown are the annualized excess return (ER), volatility (Vol), maximum drawdown (MD), the Sharpe ratio (SR), the Sortino ratio (SOR), and average asset and currency turnover, respectively. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. More details on the optimization approaches, backtest configuration, and performance metrics are provided in Chapters 3 and 4.

Comparing the performance of the Full optimizer with Overlay and Joint optimizers in Tables 5.3 to 5.5, we see that fully hedging the currency exposure performs very similarly to actively trading the currencies. Full solves the same asset-only problem as the first step of the overlay program, and the regularized Overlay and Joint optimizers achieve Sharpe ratios on par with Full in every universe. A likely reason is that the objective contains no mean term, so the currency weights are chosen only to reduce risk and not to earn the excess currency returns. Volatility in the small universe is where the Joint and Overlay outperform the Full optimizer, by up to 10%. Looking at the currency turnover, we see that active currency trading is the mechanism through which this is achieved. In the medium and large universes, the currency turnover of the Overlay and Joint optimizers is only 4% to 46% higher than that of Full, so their currency positions deviate only moderately from a full hedge, whereas in the small universe it is 1.5 to 2.3 times as high. On the other hand, the Zero optimizer keeps the full currency exposure and with it the excess currency returns that the risk-minimizing optimizers give up, and earns a higher excess return at a much higher volatility. In the small universe this gives Zero the higher Sharpe ratio for every risk measure, while in the medium and large universes the Overlay and Joint optimizers outperform it for every risk measure, by

up to 70%. This performance advantage of the risk-minimizing optimizers over Zero shows the benefits of active currency management in larger universes, where equity risk is reduced further and currency risk makes up a larger share of the portfolio's total risk.

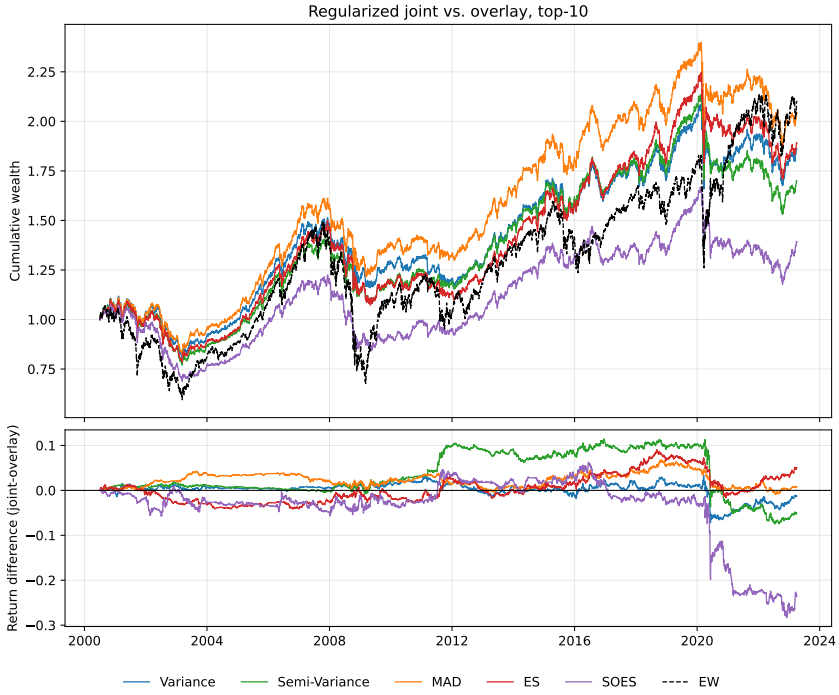


Figure 5.4: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy).

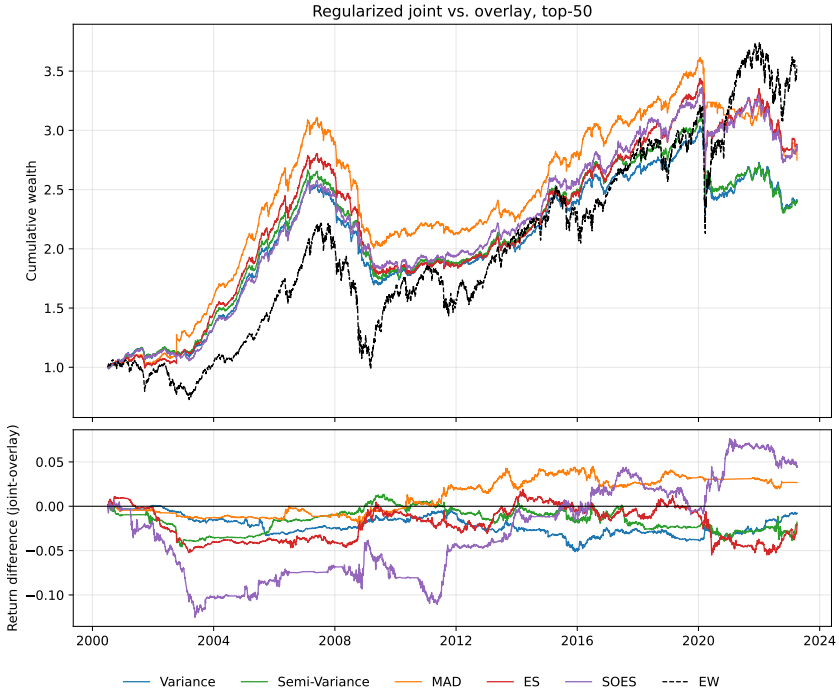


Figure 5.5: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy).

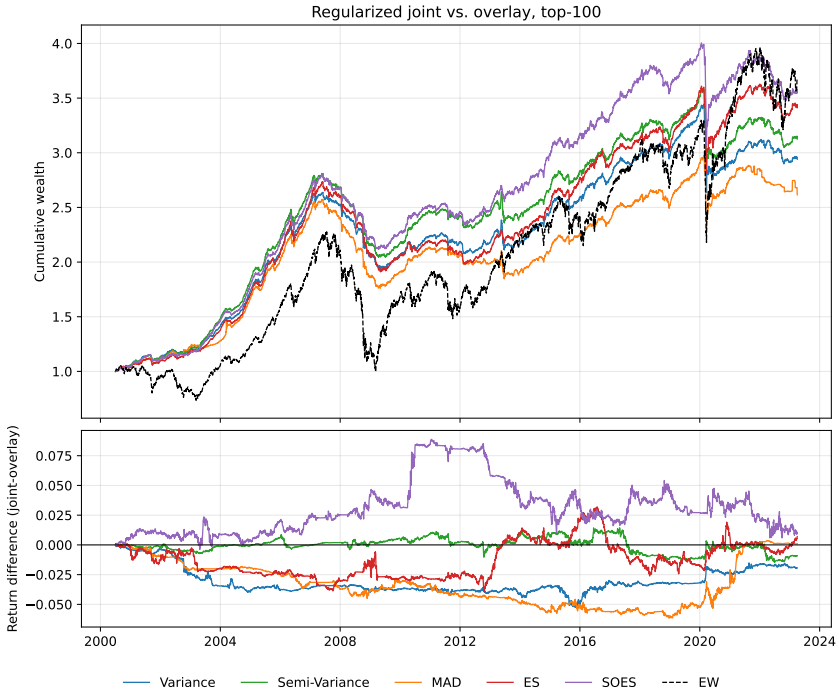


Figure 5.6: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy).

Figures 5.4 to 5.6 show the out-of-sample performance of the regularized optimizers in the small, medium and large investable universes, respectively. In each figure, the top panel shows the cumulative wealth of the joint optimizers and the equally weighted portfolio, and the bottom panel shows the cumulative sum of the daily return differences between the joint and the overlay optimizers. The portfolios are rebalanced quarterly and reported net of transaction costs. The corresponding performance metrics are reported in Tables 5.3 to 5.5.

After regularization, the performance of the Joint and Overlay optimizers remains very similar. Their Sharpe ratios in Tables 5.3 to 5.5 differ only marginally for every risk measure and universe size, with the sign of the difference changing across risk measures within each universe. The Joint optimizers again achieve a lower volatility in every measure-universe combination. The only exception being second-order Expected Shortfall in the small universe, where Overlay beats Joint on every metric. The bottom panel of Figure 5.4 shows that this happens almost entirely after the onset of the COVID-19 crisis in 2020, so it is more reflective of this period rather than a persistent difference. The bottom panels of Figures 5.5 and 5.6 show that the cumulative return differences in the medium and large universes remain small for all risk measures throughout the sample. The main effect of regularization on the comparison is on currency turnover. Without regularization, the currency turnover of Joint exceeds that of Overlay by 11% to 82%, while with regularization the two differ by at most 7%. The penalties on the currency weights thus shrink the larger currency positions of the joint program to about the size of those of the overlay program. This does not have an effect on their risk-adjusted returns, which remain very similar, with no optimizer consistently outperforming the other.

The results so far share a common feature. All optimizers minimize a risk measure without any return term in the objective, and while this delivers substantially lower volatility and drawdowns than the equally weighted portfolio, it also gives up potential excess asset and currency returns. A natural extension is therefore to add a mean term to the objective, which lets the optimizers trade off expected return against risk. A risk-return objective is also where joint optimization could be more beneficial, as the joint program can account for the

dependence structure between asset and currency returns when choosing positions that raise the expected return.

5.3 Mean-Risk Optimization

In this section, we add a mean term to the objective of the unregularized joint and overlay programs of Section 5.1. Each program now minimizes the risk measure with an added expected portfolio return scaled by $1/\gamma$, where γ is a risk aversion coefficient, as introduced in Section 3.2. In the overlay program, the asset step includes the expected asset returns and the currency step the expected excess currency returns. The expected asset returns are averaged across the stocks of each economy and shrunk toward zero, while the expected excess currency returns are set to the negative forward premium, that is, the difference between the foreign and the US interest rate. The choice of γ for each risk measure is described in **??**. All other aspects of the backtests remain unchanged. Table 5.6 reports the change in the Sharpe ratio from adding the mean term, and Table 5.7 reports the performance metrics.

Change in Sharpe Ratio from Adding a Mean Term

	Overlay			Joint		
Risk Measure	**Small**	**Medium**	**Large**	**Small**	**Medium**	**Large**
Variance	+0.23	+0.24	+0.22	+0.21	+0.24	+0.25
Semi-Variance	+0.30	+0.16	+0.12	+0.30	+0.17	+0.18
MAD	+0.18	+0.15	+0.14	+0.19	+0.21	+0.18
ES	+0.04	−0.01	0.00	+0.05	+0.05	+0.04
SOES	+0.09	+0.04	+0.02	+0.10	+0.14	+0.08
Deviation measures	+0.24	+0.18	+0.16	+0.23	+0.21	+0.20
Tail measures	+0.07	+0.02	+0.01	+0.08	+0.10	+0.06

Table 5.6: Change in the out-of-sample Sharpe ratio of the Overlay and Joint optimizers from adding a mean term

Table 5.6 reports the change in the out-of-sample Sharpe ratio from adding a mean term to the objective, computed as the Sharpe ratio of each mean-risk optimizer in Table 5.7 minus that of the unregularized version in Table 5.1, for the Overlay and Joint optimizers in the small, medium and large investable universes. Rows correspond to the risk measure in the objective, that is, variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES) and second-order Expected Shortfall (SOES), and the last two rows report the average change across the deviation measures (variance, semi-variance and MAD) and across the tail measures (ES and SOES).

Looking at Table 5.6, we see that adding a mean term to the objective raises the Sharpe ratios of the Overlay and Joint optimizers in 28 of the 30 combinations of risk measure and universe size. Unlike regularization, which was most beneficial in the large universe, the addition of a mean term improves the Sharpe ratios by a similar amount in all three universes. Minimizing risk alone ignores expected returns, and the mean term recovers part of the excess return the optimizers otherwise give up. Though we can see that the improvement is very uneven across risk measures. The Sharpe ratios of the deviation measures rise by 0.16 to 0.24 on average, while those of the tail measures rise by at most 0.10.

This split is in line with how the risk measures handle the mean of the portfolio return. Expected Shortfall and second-order Expected Shortfall are translation equivariant, so minimizing them already rewards portfolios with a higher mean. The two tail measures therefore carry an implicit mean term, which the explicit one only strengthens. The deviation measures are

invariant to a shift in the portfolio return, so for them the addition of a mean term introduces new information in the optimization problem. The average currency turnover is consistent with this interpretation. In the medium and large universes, the mean term raises the currency turnover of the variance and semi-variance optimizers to two to three times its previous level, while for the tail-measure optimizers it remains stable around its previous level. We analyze the implicit risk-return tradeoff of the tail measures in more detail in Section 6.2.

Mean-Risk Backtest Performance across Universe Sizes

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	1.77	3.21	3.13	3.75	3.75	2.81	3.05	0.96	1.01	1.51	1.03
Vol (%)	13.96	8.56	8.40	9.85	9.81	8.31	8.13	8.20	7.95	9.19	8.75
MD (%)	54.77	26.41	25.91	30.52	30.17	26.52	25.58	29.58	27.57	33.77	37.06
SR	0.13	0.37	0.37	0.38	0.38	0.34	0.37	0.12	0.13	0.16	0.12
SOR	0.17	0.51	0.51	0.53	0.53	0.46	0.51	0.16	0.17	0.23	0.16
Turnover^a	0.30	0.94	0.92	1.17	1.22	1.13	1.13	1.30	1.28	1.39	1.48
Turnover^c	0.92	2.36	2.41	2.65	2.61	2.14	2.25	2.15	2.35	2.25	2.42
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.09	4.54	4.66	4.78	5.21	3.59	3.82	3.15	3.26	2.67	3.54
Vol (%)	12.50	6.37	6.33	7.79	7.95	5.68	5.57	5.68	5.58	5.89	5.70
MD (%)	55.75	37.66	36.85	40.93	41.50	36.19	37.92	39.52	38.75	34.11	32.47
SR	0.33	0.71	0.74	0.61	0.66	0.63	0.69	0.55	0.58	0.45	0.62
SOR	0.44	0.96	1.00	0.83	0.90	0.87	0.95	0.75	0.79	0.61	0.86
Turnover^a	0.27	1.17	1.21	1.43	1.52	1.64	1.69	1.78	1.80	2.13	2.25
Turnover^c	0.92	2.62	2.71	3.15	3.10	1.60	2.01	0.99	1.72	1.03	1.92
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		**Semi-Variance**		**MAD**		**ES**		**SOES**	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.26	5.02	5.26	5.57	6.39	3.20	3.25	3.30	3.66	2.51	2.90
Vol (%)	11.66	5.73	5.76	7.36	7.59	4.79	4.68	4.59	4.57	4.74	4.72
MD (%)	56.26	31.32	30.62	37.45	37.04	31.43	33.29	31.65	30.47	31.65	29.57
SR	0.37	0.87	0.91	0.76	0.84	0.67	0.70	0.72	0.80	0.53	0.61
SOR	0.49	1.17	1.23	1.03	1.16	0.89	0.91	0.95	1.07	0.70	0.82
Turnover^a	0.28	1.24	1.29	1.56	1.65	1.74	1.85	1.89	1.92	2.14	2.20
Turnover^c	0.91	2.88	2.95	3.40	3.36	1.37	1.94	0.97	1.49	1.00	1.80

Table 5.7: Performance metrics of the mean-risk optimizers in the small, medium and large investable universes

Table 5.7 reports out-of-sample performance metrics for the mean-risk optimizers in the three investable universes, with portfolios rebalanced quarterly. Panel A covers the small, Panel B the medium and Panel C the large investable universe. Each panel reports the unregularized sequential currency overlay (Overlay) and joint (Joint) optimizers with an added mean term, together with the equally weighted portfolio (EW) as a benchmark. Columns are grouped by the risk measure, that is, variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES) and second-order Expected Shortfall (SOES). The realized out-of-sample performance metrics shown are the annualized excess return (ER), volatility (Vol), maximum drawdown (MD), the Sharpe ratio (SR), the Sortino ratio (SOR), and average asset and currency turnover, respectively. The sample period runs from June 1990 to June 2023, and the first out-of-sample portfolio is formed in June 2000. More details on the optimization approaches, backtest configuration, and performance metrics are provided in Chapters 3 and 4.

As we can see in Table 5.7, the higher Sharpe ratios come from substantially higher excess returns achieved by the mean-risk optimizers. The excess returns are increased by 1.9 to 3.6 percentage points for the variance and semi-variance optimizers and by 0.8 to 1.7 percentage points for the mean absolute deviation optimizers, but by at most 0.8 percentage points for the tail measures. This comes at the cost of an increase in volatility, which rises by 0.6 to 3.4 percentage points for the variance and semi-variance optimizers, while for mean absolute deviation and the tail measures it changes by less than half a percentage point. Even with the increase, the volatility of the variance and semi-variance optimizers remains at between 50% and 70% of that of the equally weighted portfolio, and the maximum drawdowns of all mean-risk optimizers stay below 70% of that of the equally weighted portfolio. The mean term thus raises returns without giving up the risk reduction that distinguishes the optimizers from the equally weighted portfolio. As a result, every mean-risk optimizer outperforms the equally weighted portfolio on a risk-adjusted basis in the medium and large universes. In the small universe, the deviation measures reach Sharpe ratios of 0.34 to 0.38 against 0.13 for the equally weighted portfolio, while the tail measures perform similarly at 0.12 to 0.16. For the deviation measures, the underperformance of the pure risk-minimizing optimizers in the small universe is therefore not a failure of risk reduction, but a consequence of ignoring expected returns. The Sortino ratios show the same pattern, so a comparison focused on downside risk does not change the conclusion.

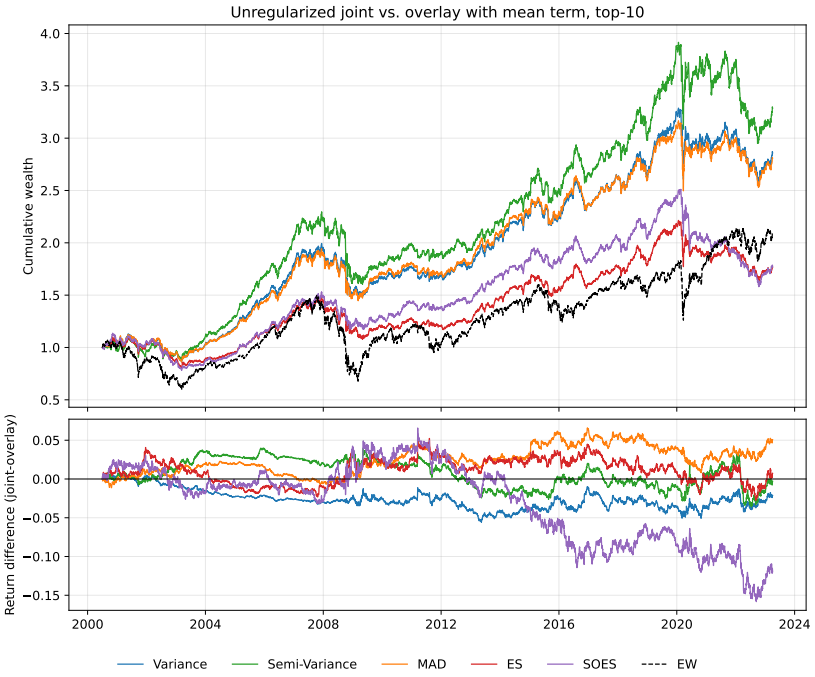


Figure 5.7: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy).

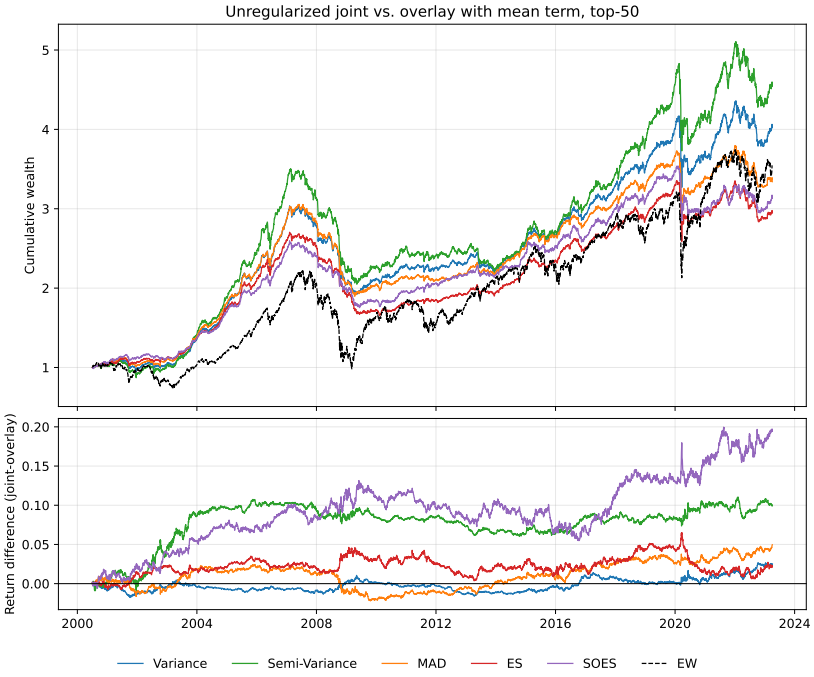


Figure 5.8: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy).

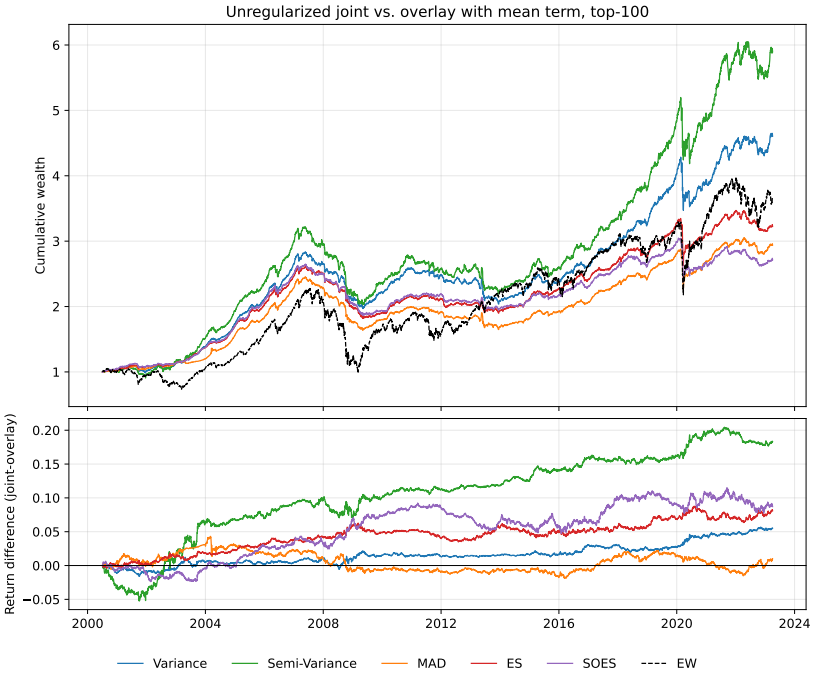


Figure 5.9: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy).

Figures 5.7 to 5.9 show the out-of-sample performance of the mean-risk optimizers in the small, medium and large investable universes, respectively. In each figure, the top panel shows the cumulative wealth of the joint optimizers and the equally weighted portfolio, and the bottom panel shows the cumulative sum of the daily return differences between the joint and the overlay optimizers. The portfolios are rebalanced quarterly and reported net of transaction costs. The corresponding performance metrics are reported in Panels A, B and C of Table 5.7, respectively.

The addition of a mean term also changes the comparison between the Joint and Overlay optimizers. In the previous sections, neither optimization approach consistently outperformed the other. Now Joint attains a higher Sharpe ratio than Overlay for every risk measure in the medium and large universes, by 0.03 to 0.17 and by about 0.07 on average. The Sortino ratios confirm this, with Joint performing better, by 0.02 to 0.25. The bottom panels of Figures 5.8 and 5.9 show that this advantage builds up gradually over the sample rather than in a single period, with the cumulative return difference reaching about 0.18 for semi-variance in the large universe and 0.19 for second-order Expected Shortfall in the medium universe by the end of the sample. At the same time, Joint attains a lower volatility than Overlay for all risk measures, by up to 0.44 percentage points. The lower volatility of the Joint optimizers is a persistent finding across all backtests in Sections 5.1 and 5.2. Intuitively, the joint program can choose assets and currencies that offset each other's risk, while the overlay program picks its assets before considering the currencies. Without a mean term, this flexibility is used only to reduce risk, which lowers volatility but leaves returns largely unchanged. With a mean term, the joint program can also weigh the expected returns of assets and currencies against each other, keeping currency exposure where it is expected to pay and hedging it where it is not.

5.4 Conclusion

All optimizers considered in this chapter achieve a much lower volatility and smaller drawdowns than the equally weighted portfolio. Without a mean term, the excess return of the Joint and Overlay optimizers is also lower. They minimize risk without any return term in the objective and have a lower exposure to the common movement of the equity markets, so they earn a smaller share of the equity risk premium. In the medium and large universes, the optimizers achieve a higher Sharpe ratio than the equally weighted portfolio, but in the small universe they do not outperform it consistently. With fewer stocks to choose from, the optimizers lose more excess return than they reduce volatility, while a larger universe offers more low-risk stocks, and the reduction in volatility outweighs the lost return.

Regularization increases the Sharpe ratios, especially in the large universe. The penalties limit the extreme positions that the unregularized programs fit to noise in the lookback window, which matters more when the number of assets is large relative to the length of the window. The regularized Joint and Overlay optimizers perform on par with the Full optimizer, which keeps the currency exposure fully hedged. Without a mean term in the objective, actively trading the currencies therefore adds little compared to full hedging in terms of returns, likely because the currency weights are then chosen only to reduce risk and not to earn the excess currency returns. Compared with the Zero optimizer, which keeps the full currency exposure, managing the currencies pays off in the medium and large universes, where equity risk is reduced further and currency risk makes up a larger share of the portfolio's total risk.

Adding a mean term to the objective raises the Sharpe ratios in almost every case and by a similar amount in all three universes. The gain comes from higher excess returns, while the volatility and drawdowns stay well below those of the equally weighted portfolio. The most significant improvement is for the deviation measures. Since they are invariant to a shift in the portfolio return, minimizing them ignores expected returns entirely, and the mean term adds information the optimizers did not use before. Expected Shortfall and second-order Expected Shortfall are translation equivariant and already reward a higher mean, so the explicit mean term changes them much less.

The Joint and Overlay optimizers perform very similarly when they only minimize risk. The Joint optimizer achieves a lower volatility and, without regularization, a higher currency

turnover. With regularization, the currency turnover decreases to about the size of those of the Overlay. In both cases, neither approach achieves consistently higher risk-adjusted returns. This changes when a mean term is added. The Joint optimizer then has higher Sharpe and Sortino ratios than the Overlay for every risk measure in the medium and large universes, and this advantage is consistent over the sample. Once expected returns enter the objective, the joint program can optimize the expected returns of assets and currencies against each other, which the overlay program cannot do since it fixes the asset weights first. This is the setting in which active currency management adds value, consistent with the result that it adds little when the currency weights are only optimized to reduce risk.

The lower volatility of the Joint optimizer is the most consistent difference between the two approaches. We would expect this, since both programs minimize risk and the joint program can always choose the same portfolio as the overlay program. Intuitively, the joint program can choose assets and currencies that offset each other's risk, while the overlay program picks its assets before considering the currencies, and this matters most in the small universe, where the number of assets is small relative to the number of currencies. In the next chapter, we therefore focus on comparing the optimizers in terms of reducing the risk measure they are optimizing for.

Chapter 6

Joint versus Sequential Optimization

In Chapter 5 we showed that the joint and overlay programs achieve similar returns, and that the most consistent difference between them is the lower realized volatility of the joint optimizers. In this chapter, we analyze the performance of the joint and overlay programs in terms of risk minimization. We first show in Section 6.1 that the joint program has an advantage over the overlay program by construction in terms of minimizing its target risk measure, since the overlay program's solution is feasible for the joint program. We define the in-sample and out-of-sample gaps between the two programs and investigate the realized advantage of joint optimization across the five risk measures for investable universes of different sizes. The results, as in Chapter 5, separate the risk measures into two groups, deviation and tail measures. In Section 6.2 we look at the tail measures more closely. We perform the same gap analysis across different confidence levels and show that the in-sample advantage of joint optimization grows with the tail depth while the out-of-sample advantage does not. This identifies a tail-fitting mechanism that does not generalize. We then transform the tail measures into deviation measures by de-meaning them and show that the in-sample advantage begins to fall with universe size, which identifies an implicit risk-return tradeoff as a second cause of the tail measures in-sample advantage. Finally, in Section 6.3 we focus on the deviation measures and argue, with the two mechanisms from Section 6.2, why a larger percentage of the in-sample advantage of joint optimization is realized out of sample.

6.1 Advantage of Joint Optimization for Risk Minimization

In sample, joint optimization provides a benefit over sequential optimization by construction when it comes to minimizing the target risk measure. We write the weight vector as $\boldsymbol{\theta} = (\boldsymbol{x}, \boldsymbol{\psi})$ and let $\Theta = \mathcal{X} \times \mathbb{R}^M$ denote the feasible set defined by the budget and leverage constraints as in Chapter 3. For a rebalancing period $t \in \{1, \ldots, T\}$, let ρ_t^{IS} be the in-sample risk over the lookback window and ρ_t^{OOS} the out-of-sample realized risk over the next holding period. Let $\hat{\boldsymbol{\theta}}_t$ denote the weight vector an optimizer returns on the lookback window, and $\rho_t^{\text{OOS}}(\hat{\boldsymbol{\theta}}_t)$ and $\rho_t^{\text{IS}}(\hat{\boldsymbol{\theta}}_t)$ are the risk these weight vectors realize. The two optimization programs are written as

$$\hat{\boldsymbol{\theta}}_{J,t} \in \operatorname*{arg\,min}_{\boldsymbol{\theta} \in \Theta} \rho_t^{\text{IS}}(\boldsymbol{\theta}),$$

and

$$\hat{\boldsymbol{x}}_{O,t} \in \operatorname*{arg\,min}_{\boldsymbol{x} \in \mathcal{X}} \rho_{A,t}^{\text{IS}}(\boldsymbol{x}), \quad \hat{\boldsymbol{\psi}}_{O,t} \in \operatorname*{arg\,min}_{\boldsymbol{\psi} \in \mathbb{R}^M} \rho_t^{\text{IS}}(\hat{\boldsymbol{x}}_{O,t}, \boldsymbol{\psi}),$$

where $\rho_{A,t}^{\text{IS}}(\boldsymbol{x}) \coloneqq \rho_t^{\text{IS}}(\boldsymbol{x}, \mathbf{0})$ is the in-sample risk of the fully hedged asset portfolio, and $\hat{\boldsymbol{\theta}}_{i,t} = (\hat{\boldsymbol{x}}_{i,t}, \hat{\boldsymbol{\psi}}_{i,t})$, $i \in \{J, O\}$. The subscript J denotes the joint optimizer and O the sequential

overlay optimizer. The overlay optimizer is a restricted minimizer. It first minimizes over $\boldsymbol{x}$ against a different objective, then over $\boldsymbol{\psi}$ with $\hat{\boldsymbol{x}}_{O,t}$ held fixed.

The overlay optimizer's output satisfies the joint program's constraints, so $\hat{\boldsymbol{\theta}}_{O,t} \in \Theta$, and its second step is a partial minimization of the same ρ_t^{IS}. Hence

$$\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t}) = \min_{\boldsymbol{\theta}\in\Theta} \rho_t^{\mathrm{IS}}(\boldsymbol{\theta}) \;\leq\; \min_{\boldsymbol{\psi}\in\mathbb{R}^M} \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{x}}_{O,t}, \boldsymbol{\psi}) = \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}).$$

We define the in-sample gap between the two optimizers as

$$g_t^{\mathrm{IS}} \coloneqq \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t}) \;\geq\; 0.$$

The in-sample gap is non-negative and measures the advantage of joint optimization over sequential optimization in the lookback window. Naturally, the question we are interested in is whether this advantage is realized out of sample.

We define the out-of-sample error of a fitted $\hat{\boldsymbol{\theta}}_t$ as

$$\epsilon_t(\hat{\boldsymbol{\theta}}_t) \coloneqq \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_t) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_t),$$

the amount by which realized risk in the holding quarter exceeds the risk the optimizer achieved in sample.[1] Then the out-of-sample gap is defined as

$$\begin{aligned} g_t^{\mathrm{OOS}} &\coloneqq \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{J,t}) \\ &= \underbrace{\left[\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t})\right]}_{g_t^{\mathrm{IS}} \geq 0} - \underbrace{\left[\epsilon_t(\hat{\boldsymbol{\theta}}_{J,t}) - \epsilon_t(\hat{\boldsymbol{\theta}}_{O,t})\right]}_{\Delta\epsilon_t}. \end{aligned}$$

The difference between the out-of-sample errors $\Delta\epsilon_t$ can be either positive or negative. For the variance, the in-sample risk of an optimized portfolio is known to understate its true risk, and the more weights are fitted relative to the sample size, the larger this understatement becomes [**?**]. Since the joint program fits all of its weights to the same in-sample objective, we expect it to adapt more closely to the noise in the lookback window than the overlay program, and hence $\Delta\epsilon_t$ to be typically positive. If $\Delta\epsilon_t$ exceeds g_t^{IS}, the overlay program could in some cases realize lower risk than the joint program out of sample. We also expect $\Delta\epsilon_t$ to be measure-dependent. Tail measures, in our case Expected Shortfall and second-order Expected Shortfall, are estimated from a smaller fraction of the sample, whereas the deviation measures, variance and mean absolute deviation use every observation, and semi-variance those below the mean. For a given sample size, we therefore expect $\Delta\epsilon_t$ to be larger and more volatile for the tail measures.

We perform the following gap analysis to investigate the in-sample and out-of-sample performance of the joint and overlay programs with respect to the risk measure they are minimizing. The in-sample gap is guaranteed to be non-negative only when each program minimizes the risk measure it is scored on, so we consider the unregularized formulation from Chapter 3. At each of the $T = 91$ quarterly rebalancing dates the joint and the overlay optimizers, for a given risk measure, are solved on the same ten-year lookback window, and under the same budget and leverage constraints. The resulting weights are then held fixed and scored twice, once on the lookback window itself and once on the following holding quarter, which gives ρ_t^{IS} and ρ_t^{OOS} for each of the two programs. Both gaps are reported relative to the risk of the overlay optimizer

[1] This quantity is the portfolio analogue of the optimism of the training error, see for example Section 7.4 in [**?**].

$$\tilde{g}_t^{\mathrm{IS}} := \frac{g_t^{\mathrm{IS}}}{\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t})} \qquad \text{and} \qquad \tilde{g}_t^{\mathrm{OOS}} := \frac{g_t^{\mathrm{OOS}}}{\rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{O,t})},$$

so they are to be interpreted as the risk reduction achieved by the joint optimizer over the overlay optimizer, expressed as a percentage of the latter's risk. For each risk measure and universe size we report the mean of the relative gaps across the rebalancing dates together with a 95% bootstrap confidence interval based on 10,000 resamples.

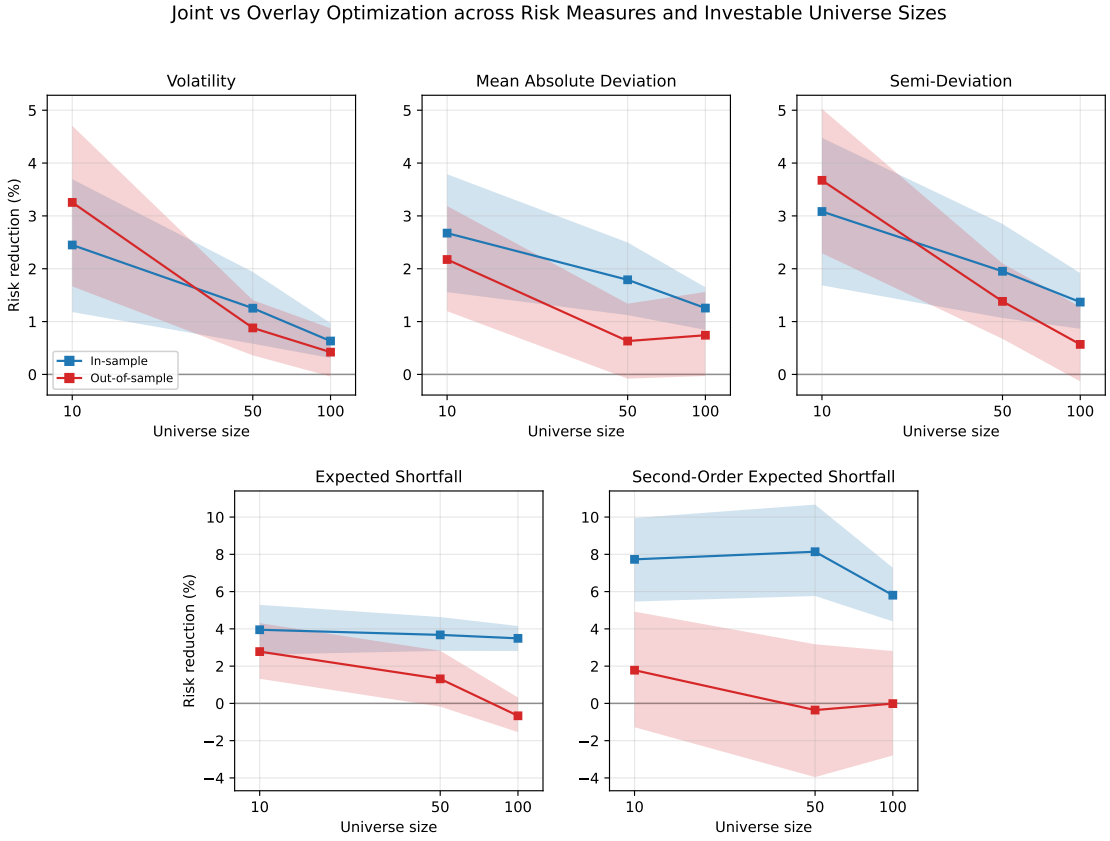


Figure 6.1: In-sample and out-of-sample relative gaps between the unregularized joint and overlay optimizers, by risk measure and universe size.

Figure 6.1 reports the mean in-sample and out-of-sample relative gaps between the unregularized joint and overlay optimizers, expressed as a percentage of the overlay optimizer's risk. Each panel reports one risk measure. The horizontal axis is the size of the investable universe and the vertical axis the risk reduction the joint program achieves over the overlay program. Squares mark the mean across the $T = 91$ quarterly rebalancing dates, in blue for the in-sample gap $\tilde{g}_t^{\mathrm{IS}}$ and in red for the realized gap $\tilde{g}_t^{\mathrm{OOS}}$. Shaded areas mark the 95% bootstrapped confidence intervals.

We see in Figure 6.1 that the in-sample gap of joint optimization is positive for all risk measures across all universe sizes, and that it is largest for the two considered tail measures. The out-of-sample gap is smaller than the in-sample gap for almost all risk measures and universe sizes, the only exception being the variance and semi-variance measures in the small universe. Out of sample, we see a negative dependence on the size of the investable universe. In the small universe the realized out-of-sample advantage of joint optimization is in the range between 1.78% and 3.67%, with second-order Expected Shortfall having the smallest benefit and semi-variance having the largest. In the medium universe, the advantage drops to zero for second-order Expected Shortfall, and remains around 1% for the other measures, and in the large universe the advantage for Expected Shortfall drops slightly below zero while the remaining measures stay below 1%.

The in-sample series separates the two groups of risk measures the same way as in Chapter 5, that is, whether they are deviation or tail measures. The in-sample advantage falls as the universe grows, from 2.45% to 0.63% for variance, from 2.67% to 1.26% for mean absolute deviation and from 3.08% to 1.37% for semi-variance, whereas for the two tail measures it remains larger and more stable, from 3.95% to 3.49% for Expected Shortfall and from 7.73% to 5.81% for second-order Expected Shortfall. For the deviation measures the in-sample and out-of-sample series fall together, whereas for the tail measures, the out-of-sample series falls to zero, and the in-sample series remains large and positive. We visualize the difference between the two series in Figure 6.2.

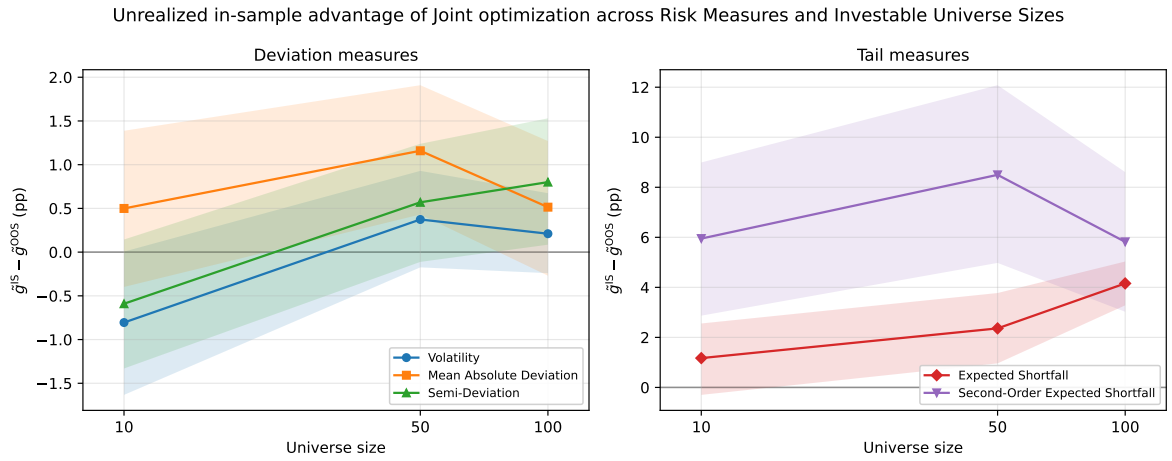


Figure 6.2: Unrealized in-sample advantage of unregularized joint optimization, $\tilde{g}_t^{\text{IS}} - \tilde{g}_t^{\text{OOS}}$, by risk measure and universe size.

Figure 6.2 reports $\tilde{g}_t^{\text{IS}} - \tilde{g}_t^{\text{OOS}}$ for the unregularized optimizers, the percentage advantage the joint optimizer holds in sample minus the percentage advantage it realizes out of sample. For every risk measure, we mark the mean difference across the 91 quarterly rebalancing dates, with shaded areas marking the 95% bootstrap intervals obtained the same way as in Figure 6.1. Being a difference of two ratios, the quantity reported is measured in percentage points. The deviation and the tail measures are shown on separate panels, with independent vertical scales, because the latter are an order of magnitude larger.

The first thing we notice in Figure 6.2 is that the bootstrapped confidence intervals for the two tail measures are wider than those for the three deviation measures, averaging 4.37 against 1.41 percentage points. This is expected, since the tail measures are estimated on a smaller fraction of the sample. The second thing is the size of the unrealized advantage itself, which is between 1.17 and 8.50 percentage points for the two tail measures while the three deviation measures are between -0.81 and 1.16 percentage points. This, together with the in-sample series of Figure 6.1 (which for the tail measures stays between 3.49% and 8.14% at every universe size), shows that the tail measures have a larger in-sample advantage coming from joint optimization, that is not realized out of sample, whereas for the deviation measures the advantage that is achieved in sample is largely realized out of sample but drops with universe size. The two groups produce different results, and we analyze the mechanism behind them separately in Sections 6.2 and 6.3.

6.2 Tail Measures

6.2.1 Level Analysis

To investigate the mechanism behind the in-sample advantage of joint optimization for the two tail measures, Expected Shortfall and second-order Expected Shortfall, we perform the same gap analysis as in Section 6.1 across a range of confidence levels. We report the results in Figures 6.3 and 6.4.

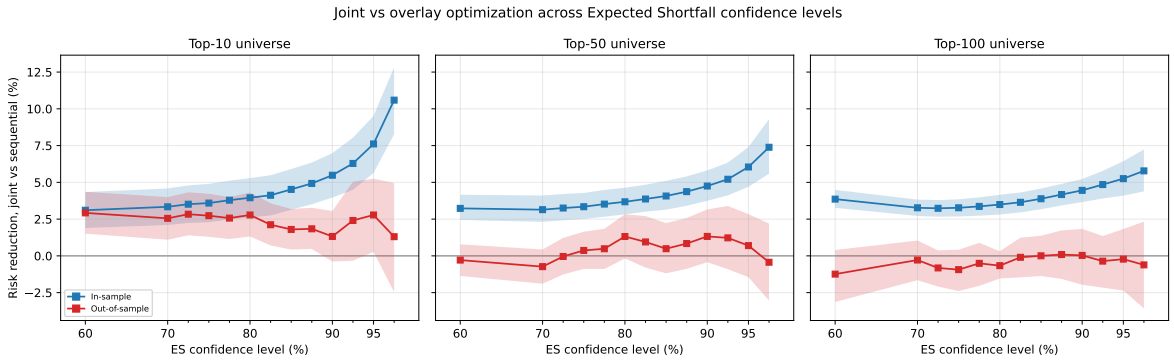


Figure 6.3: In-sample and out-of-sample relative gaps between the unregularized joint and overlay Expected Shortfall optimizers, by confidence level and universe size.

Figure 6.3 reports the mean in-sample and out-of-sample relative gaps between the unregularized joint and overlay Expected Shortfall optimizers, expressed as a percentage of the overlay optimizer's risk, with the confidence level on the horizontal axis and one panel per universe size. Squares mark the mean across the $T = 91$ quarterly rebalancing dates, in blue for the in-sample gap $\tilde{g}_t^{\mathrm{IS}}$ and in red for the realized gap $\tilde{g}_t^{\mathrm{OOS}}$. Shaded areas mark the 95% bootstrapped confidence intervals. The figures cover thirteen confidence levels from 60% to 97.5%.

We see in Figure 6.3 that the in-sample advantage of joint optimization for minimizing Expected Shortfall grows with the confidence level and the growth is more pronounced in the smaller universe. The out-of-sample advantage is much smaller and does not rise with the confidence level at any universe size. In the small universe, the in-sample advantage rises from 3.11% at confidence level 60% to 10.59% at level 97.5%, while the out-of-sample advantage stays in the 1.30 to 2.92% range. In the larger universes the in-sample advantage grows at a lower rate, going from 3.23% to 7.39% in the medium and from 3.86% to 5.78% in the large universe, while the realized advantage never exceeds 1.32% and 0.09% respectively.

Expected Shortfall at confidence level α is the average loss over the worst $1 - \alpha$ fraction of the lookback window, which is roughly 1,000 observations at 60% and 63 at 97.5%. Which observations fall in the tail depends largely on the portfolio weights. The joint program chooses the asset and currency weights together, so it can select assets whose worst days the currency positions can offset. The overlay program fixes the asset weights of the fully hedged portfolio, and its currency step can adjust M weights against a largely fixed set of the worst performing days. As we increase the confidence level, the number of observations in the tail shrinks, so the joint program's currency weights can cover a larger share of the worst performing days, and the in-sample advantage grows. The growth is largest in the small universe, where the currency weights are a larger share of all the decision variables. In larger universes, the overlay program's asset step has more weights to diversify, so the joint program's advantage decreases. Since the weights are fitted to specific worst performing days of the lookback window that do not necessarily occur in the next period, the mechanism generalizes poorly out of sample.



Figure 6.4: In-sample and out-of-sample relative gaps between the unregularized joint and overlay second-order Expected Shortfall optimizers, by confidence level and universe size.

Figure 6.4 reports the mean in-sample and out-of-sample relative gaps between the unregularized joint and overlay second-order Expected Shortfall optimizers, expressed as a percentage of the overlay optimizer's risk, with the confidence level on the horizontal axis and one panel per universe size. Squares mark the mean across the $T = 91$ quarterly rebalancing dates, in blue for the in-sample gap $\tilde{g}_t^{\mathrm{IS}}$ and in red for the realized gap $\tilde{g}_t^{\mathrm{OOS}}$. Shaded areas mark the 95% bootstrapped confidence intervals. The figures cover thirteen confidence levels from 60% to 97.5%.

In Figure 6.4 we see the same pattern for second-order Expected Shortfall. The in-sample advantage of joint optimization grows with the confidence level and the growth is faster in the smaller universe. We also observe that above certain confidence levels, the in-sample advantage becomes constant. The out-of-sample advantage is much smaller and does not rise with the confidence level at any universe size. In the small universe, the in-sample advantage rises from 4.00% at level 60% to 16.28% at level 92.5% and from there remains constant. The out-of-sample advantage stays in the -0.45 to 3.88% range. In the larger universes the in-sample advantage grows at a lower rate and becomes constant at lower confidence levels, going from 3.44% to 8.52% in the medium and from 3.83% to 5.81% in the large universe, while the out-of-sample advantage fluctuates around zero.

Second-order Expected Shortfall squares the tail observations, so more weight is placed on the most extreme observations. At level α it therefore behaves like Expected Shortfall at a higher confidence level, which is why the in-sample advantage is larger at every confidence level and every universe size. As we increase the confidence level, the squared weighting concentrates on a smaller and more stable set of the worst performing days, and above a certain level, raising the confidence level further no longer changes which observations dominate the objective, and the in-sample series becomes constant. In larger universes this happens at lower confidence levels, since the overlay program's asset step has more weights to fit those days. Out of sample, the same reasoning as for Expected Shortfall applies, and the realized advantage does not rise with the confidence level at any universe size and is only present in the small universe.

6.2.2 De-meaning the Tail Measures

As discussed in Chapter 3, Expected Shortfall and second-order Expected Shortfall are coherent (strictly expectation bounded) risk measures rather than deviation measures. Unlike variance, mean absolute deviation and semi-variance, they are not invariant to translation, so a portfolio with a higher mean return is scored favorably to one with a lower mean return, everything else being equal. The objectives therefore carry an implicit risk-return tradeoff that the three deviation measures do not. We can transform a coherent strictly expectation bounded risk measure $\mathcal{R}$ into a deviation measure $\mathcal{D}$ via a simple de-meaning operation. More precisely

Theorem D.3 *Let $\mathcal{R}$ be a coherent strictly expectation bounded risk measure, and let $\mathcal{D}$ be the functional defined by*

$$\mathcal{D}(X) \coloneqq \mathcal{R}(X - \mathbb{E}[X]).$$

Then $\mathcal{D}$ is a deviation measure.

The proof of Theorem D.3 and further details can be found in Appendix D. We perform the same gap analysis as in Section 6.1 on the de-meaned Expected Shortfall and second-order Expected Shortfall optimization programs, and report the results in Figure 6.5.

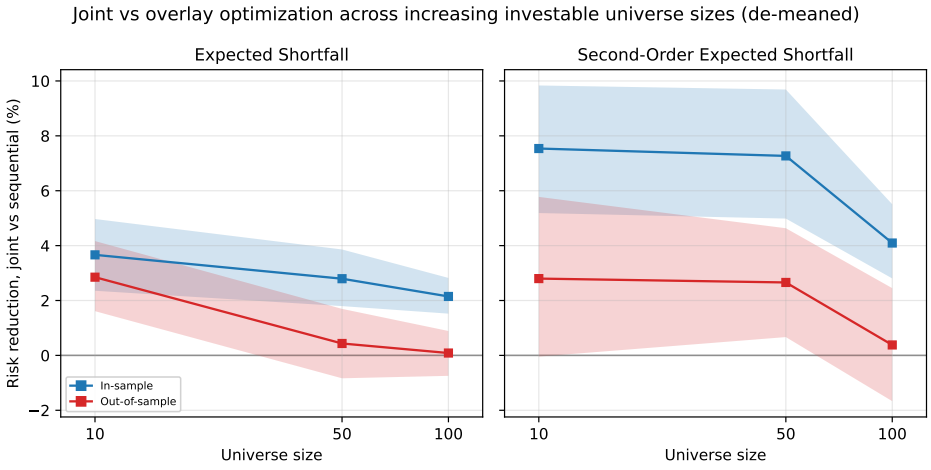


Figure 6.5: In-sample and out-of-sample relative gaps between the unregularized joint and overlay de-meaned Expected Shortfall and second-order Expected Shortfall optimizers, by universe size.

Figure 6.5 reports the mean in-sample and out-of-sample relative gaps between the unregularized joint and overlay de-meaned Expected Shortfall and second-order Expected Shortfall optimizers, expressed as a percentage of the overlay optimizer's risk, with the universe size on the horizontal axis. Squares mark the mean across the $T = 91$ quarterly rebalancing dates, in blue for the in-sample gap $\tilde{g}_t^{\mathrm{IS}}$ and in red for the realized gap $\tilde{g}_t^{\mathrm{OOS}}$. Shaded areas mark the 95% bootstrapped confidence intervals.

We see in Figure 6.5 that de-meaning the optimizers steepens the decline of the in-sample relative gaps. For Expected Shortfall the in-sample gap falls from 3.66% to 2.15% across the three universes, against 3.95% to 3.49% for the original tail measure, and for second-order Expected Shortfall from 7.54% to 4.09% against 7.73% to 5.81%. The drop in the gap across universes is 41.3% as opposed to 11.6% for Expected Shortfall and 45.7% as opposed to 24.8% for second-order Expected Shortfall. This indicates that the implicit risk-return tradeoff is one of the driving factors for the tail measures in-sample advantage as the universe grows. With it removed, they begin to decay in the way the deviation measures do. In the large universe the de-meaned Expected Shortfall gap is still 2.15% against, for example, 0.63% for variance, so the mean term accounts for only part of the in-sample advantage.

6.3 Deviation Measures

For the three considered deviation measures the advantage of joint optimization is smaller in sample than for the tail measures, and most of it survives out of sample. If we compare the two series from Figure 6.1 measure by measure, the realized gap $\tilde{g}^{\mathrm{OOS}}$ as a fraction of the in-sample gap $\tilde{g}^{\mathrm{IS}}$ is 133%, 70% and 67% for variance across the small, medium and large universes, 81%, 35% and 59% for mean absolute deviation, and 119%, 71% and 42% for semi-variance. For the tail measures the same ratio is 70%, 36% and −19% for Expected Shortfall and 23%, −4% and 0% for second-order Expected Shortfall.

Variance, semi-variance and mean absolute deviation are estimated from all observations in the window (half for semi-variance), so the objective is estimated more accurately and the out-of-sample gap follows the in-sample gap more closely. Expected Shortfall and its second-order variant are determined by a small fraction of the observations at the confidence levels considered, so the same gap is a noisier quantity. Figures 6.3 and 6.4 make this explicit, as the tail is made thinner the in-sample gap grows while the realized gap stays flat or decays. The deviation measures are also translation invariant, so they carry no implicit risk-return tradeoff, and the joint program cannot produce an in-sample advantage by tilting the weights towards a high realized mean. The tail measures can, and Figure 6.5 shows that removing the mean term accounts for part of the in-sample advantage across universes. So deviation measures are left with a small advantage that shrinks both in sample and out of sample as the investable universe grows.

6.4 Conclusion

Even though joint optimization does not consistently outperform in terms of returns, it has an advantage over sequential optimization by construction when it comes to minimizing the targeted risk measure. In sample, the advantage for the deviation measures falls with universe size, while for the two considered tail measures it stays stable. Out of sample, the advantage falls with universe size for all five considered risk measures.

The tail measures in-sample advantage is driven by two mechanisms. The first is a tail-fitting mechanism, which lets the joint program align the currency weights with the specific worst performing days of the lookback window, and which does not generalize out of sample. The second is an implicit risk-return tradeoff, which lets the joint program produce an in-sample advantage by tilting the weights towards a high realized mean. Once de-meaned, the tail measures in-sample advantage begins to fall with universe size, in the same way as the deviation measures. Neither of the two mechanisms is available to the deviation measures, which is why their in-sample advantage was smaller to begin with and a larger fraction of it survives out of sample.

The out-of-sample advantage of joint optimization is present only in the small universe, where the currency weights are a larger share of the decision variables. In larger universes the overlay program's asset step has enough weights to fit the objective on its own, and the joint program's advantage drops to zero. Joint optimization is therefore beneficial for risk reduction when the number of assets is small relative to the number of currencies. These results agree with the findings in Chapter 5, where the joint program's lower realized volatility was largest in the small universe.